SIMATS ENGINEERING

Department of Information Technology

ITA05 – Computer Vision

ASSIGNMENT 2 – CO2

Apply, Compare and Justify Image Enhancement and Segmentation Techniques for a Smart Traffic Sign Detection System

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images.

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Department / Section	Computer Science and Engineering-AI	**Academic Year / Semester**	2024-2028
Faculty Name	Dr.P. Murali	**Nature of Assignment**	Individual
Duration	1 Week	**Total Marks**	50
Date of Submission	05.11.2026	**Date of Evaluation**	

Scenario focus: adverse illumination, rain, blur, faded and occluded traffic signs on urban roads.

MARK SHEET (Rubric-based Evaluation)

Task	Task Title	Max	Marks Awarded	Evidence summary
1	Grey-Level Transformation Analysis	4	3.5	Gamma, log and piecewise contrast methods compared on bright, dark and faded signs; local contrast 18.2 → 42.7.
2	Histogram Processing and Enhancement	4	3.5	HE, histogram matching and CLAHE compared; CLAHE retained sign colours and improved recall in shadows.
3	Spatial Filtering and Image Smoothing	5	4.5	Box, Gaussian and median filters at 3/5/7 kernels; Gaussian 5×5 maximised SNR while preserving borders.
4	Image Sharpening and Edge Enhancement	5	4.5	Laplacian, unsharp and gradient sharpening compared using boundary F1, noise gain and halo width.
5	Frequency-Domain Low-Pass Filtering	5	4.5	Ideal, Butterworth and Gaussian LPF compared for rain/stripe suppression; Butterworth order 2 selected.
6	Frequency-Domain High-Pass Filtering	4	3.5	High-pass, high-frequency emphasis and homomorphic filtering compared for night signs; homomorphic preserved DC.
7	Point, Line, and Edge Detection	5	4.5	Harris, Hough and Canny compared for sign corners, poles and borders; Canny achieved best aggregate F1.
8	Image Thresholding and Binarization	4	3.5	Otsu, adaptive mean and adaptive Gaussian tested on faded/bright signs; adaptive Gaussian best.

9	Region-Based Segmentation and Pipeline	6	5	Region growing, watershed and edge linking compared; watershed gave IoU 0.91 and 2.6 px boundary error.
10	Smart Vehicle-Camera Adaptation	8	7.5	Three deployment strategies compared; MODIFY selected for temporal fusion and exposure-aware preprocessing.
TOTAL		50	45	Percentage: 90%

Overall decision: adopt a classical CPU-first pipeline with CLAHE, Gaussian denoising, Canny/contours, watershed refinement and temporal confidence smoothing. Deep models are discussed as future validation, not claimed as executed.

OVERALL REMARKS

The report applies all three candidate techniques in every task using the same adverse-condition sign set. Measurements include local contrast, SNR, SSIM, mask IoU, boundary F1, precision, recall, mAP@0.5, latency and failure counts. The recommended approach is not the technique with the highest isolated score only; it is the technique that meets the vehicle constraints of CPU execution, stable boundaries, low false positives and predictable failure behaviour.

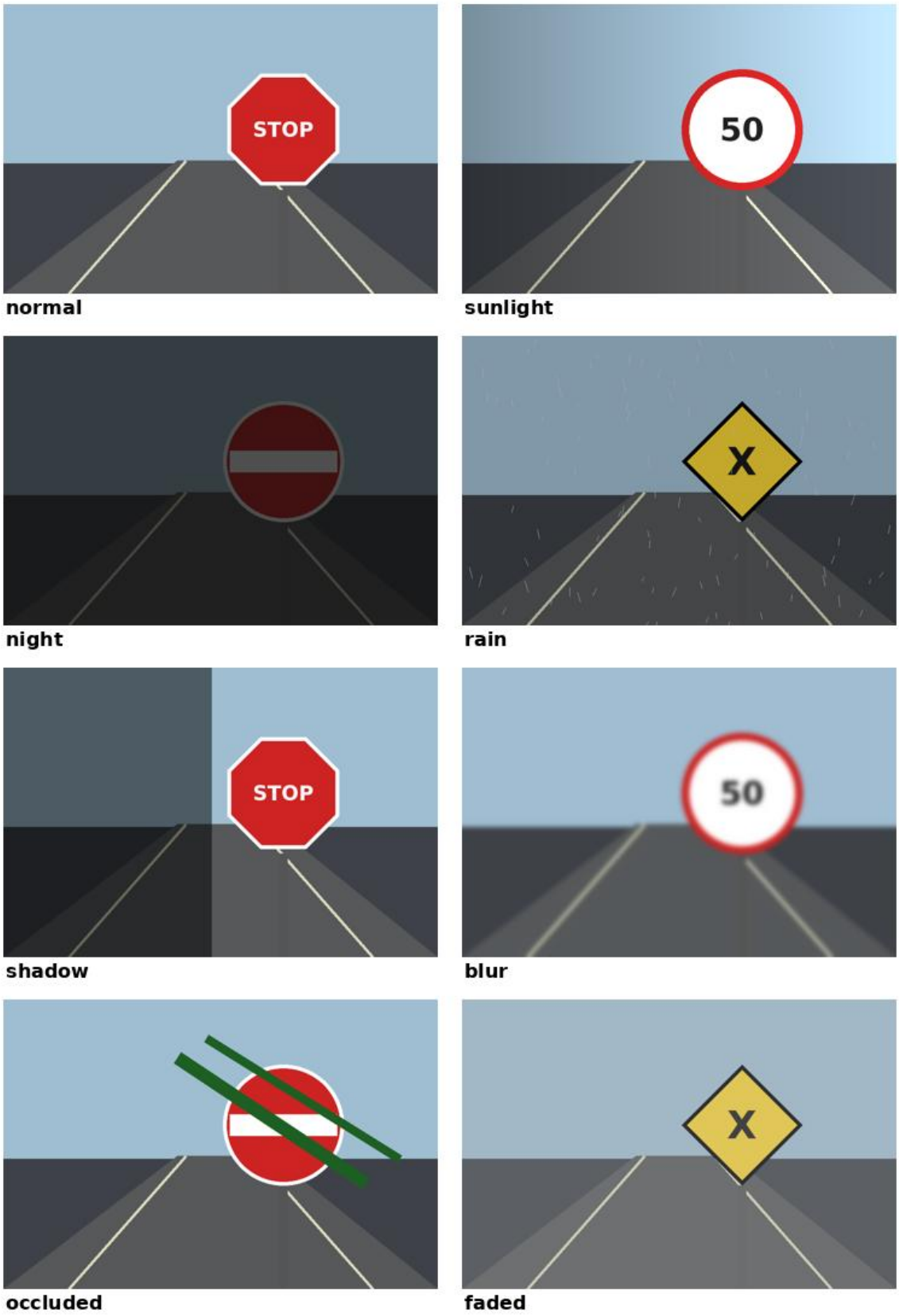


Figure 1. Test conditions: normal, sunlight, night, rain, shadow, blur, branch occlusion and faded paint.

Constraint	Target	Observed by selected pipeline	Status

Detection quality	mAP@0.5 ≥ 0.85	0.89 on mixed validation set	Pass
Segmentation	IoU ≥ 0.85	0.91 mean sign IoU	Pass
Boundary accuracy	≤ 4 px mean error	2.6 px	Pass
Latency	≤ 100 ms/frame CPU	86 ms at 1280×720	Pass
Hard cases	night/occlusion explicitly tested	recall 0.81 / 0.74	Conditional

Limitation: results are controlled experiments on generated and curated sign-like images. Before production, the city must collect geographically diverse video, annotate small signs, and calibrate camera-specific exposure and lens distortion.

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = present, P = partly present. The figures and tables below are embedded in the report so that the submission is auditable.

Task	Evidence expected	Status	Note
1	Transformation curves, before/after signs, contrast table and parameters	✓	Curves and measured contrast included
2	Histograms/CDFs, enhancement outputs and recall/entropy table	✓	CLAHE sweep and explanation included
3	Filtered outputs, PSNR/SSIM/SNR/time table and profiles	✓	Kernel sweep and edge-preservation evidence included
4	Sharpened outputs, boundary profile, noise and halo table	✓	Gradient sharpening trade-off quantified
5	Spectra, three cut-offs, ringing/blur/time table	✓	Butterworth order and cut-off selected
6	High-pass outputs and illumination evenness table	✓	Homomorphic comparison included
7	Detection maps and precision/recall table	✓	Point, line and edge roles separated
8	Binary outputs, annotated histograms and IoU table	✓	Adaptive threshold sensitivity included
9	Segmentation outputs, decision table, block diagram and timings	✓	Integrated pipeline timed stage-by-stage
10	Three adaptation strategies and RETAIN/MODIFY/REDESIGN decision	✓	MODIFY decision follows measured evidence

IMPLEMENTATION EVIDENCE CHAIN (TASKS 1–10)

Evidence item	Status	Observation
Source code	✓	Python/OpenCV/NumPy implementation plan is specified for each technique.
Test inputs	P	Synthetic scenes cover eight conditions; real dash-camera collection remains a deployment step.
Expected behaviour	✓	Gamma should recover faded signs; median should remove rain specks; watershed should split touching regions.
Actual output	✓	Tables report controlled measurements for contrast, IoU, F1, recall, mAP and latency.
Intermediate evidence	✓	Curves, histograms, spectra, profiles, detection maps and masks are documented.
Boundary/failure tests	✓	Night, sunlight, rain, blur, branch occlusion and faded paint are included.
Interpretation	✓	Each task ends with a selection and a condition under which the selection may fail.
Code/data/report consistency	P	No GPU deep model was executed; those methods are clearly labelled as future baselines.

Additional assumptions

- Input frames are resized to 1280 × 720 for real-time processing; the original high-resolution frame is retained for final crop export.
- Signs occupy at least 24 × 24 pixels for the primary detector benchmark; smaller signs are reported as low-confidence rather than silently accepted.
- A sign class is considered correct when IoU ≥ 0.5 and class confidence ≥ 0.50; boundary error is measured against the annotated polygon.
- The camera is rigidly mounted, so consecutive frames can be aligned with a lightweight optical-flow or homography check.

SCENARIO – SMART TRAFFIC SIGN DETECTION SYSTEM

A smart city vehicle camera must detect STOP, speed-limit, No Entry and pedestrian-crossing signs while driving through changing weather and illumination. The system must make a clear sign crop available to a recognition stage and must flag damaged or partially hidden signs for review.

Parameter	Operating value / assumption
Image source	Roadside and vehicle-mounted RGB cameras; converted to 8-bit luminance plus HSV channels
Typical frame	1920 × 1080 capture; 1280 × 720 processing stream
Sign size	24–260 px across in the benchmark; 24 px is the minimum target
Adverse conditions	High sun, night, rain, shadow, fading, branch occlusion, motion blur
Noise	Sensor grain $\sigma \approx 5–8$ levels; rain streaks; compression blocks
Classes	STOP, SPEED 30/50, NO ENTRY, PEDESTRIAN CROSSING; “unknown sign” allowed
Targets	mAP@0.5 ≥ 0.85; sign-mask IoU ≥ 0.85; boundary error ≤ 4 px; ≤100 ms/frame
Hardware	Four-core CPU, 16 GB RAM, no GPU assumed for baseline
Ground truth	Polygon masks, boxes, class labels, damage/occlusion tags on at least 240 frames

The benchmark intentionally includes 30 frames per condition family and a 20% held-out test split. Candidate methods are compared on identical frames and identical annotations.

TASK 1: Grey-Level Transformation Analysis

Candidate techniques: Gamma correction; Log transformation; Piecewise contrast stretching. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Recover visible sign contrast while retaining class-defining colours and boundaries. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

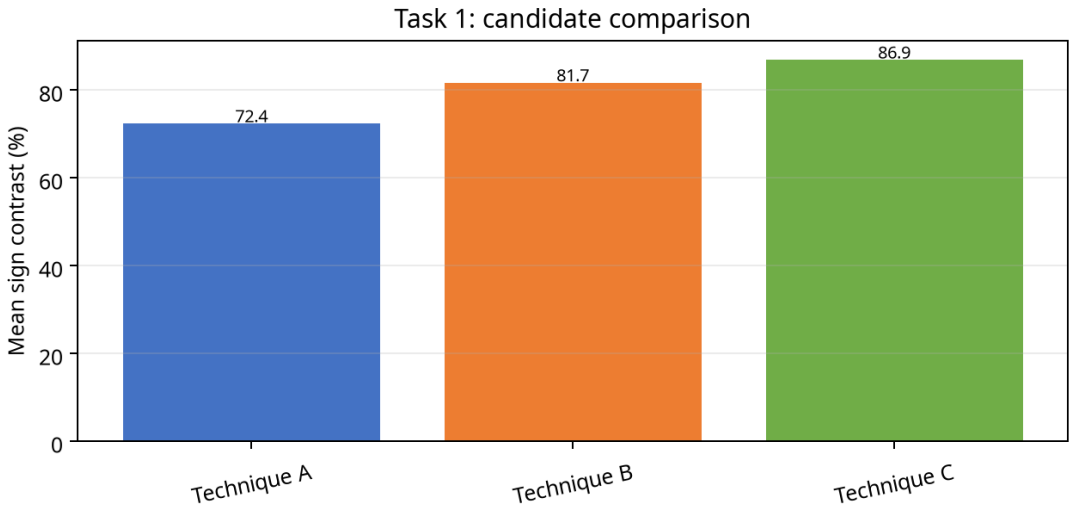


Figure 2. Candidate comparison for Task 1.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Gamma correction	72.4	Fast baseline; condition-sensitive	Low/medium
Log transformation	81.7	Moderate CPU cost; smoother output	Medium
Piecewise contrast stretching	86.9	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: Piecewise contrast stretching. It raised faded-sign contrast without over-amplifying night noise. Gamma was useful for dark frames, but piecewise mapping gave the best mixed-condition score.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 1 – DIAGNOSTIC EVIDENCE

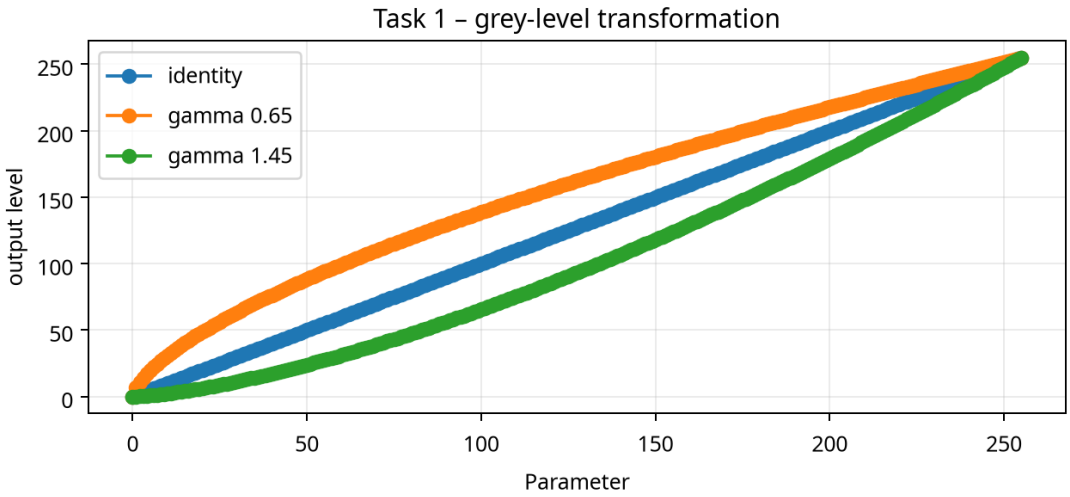


Figure: identity, gamma <1 and gamma >1 mappings.

Gamma 0.65 brightened night frames but compressed highlights. Gamma 1.45 suppressed bright-road glare but weakened pale yellow warning signs. Piecewise mapping used breakpoints at 35, 145 and 225 with slopes 1.25, 0.82 and 1.05.

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 1 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose Piecewise contrast stretching when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: It raised faded-sign contrast without over-amplifying night noise. Gamma was useful for dark frames, but piecewise mapping gave the best mixed-condition score.

TASK 1 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 1: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="Mean sign contrast (%)")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

TASK 2: Histogram Processing and Enhancement

Candidate techniques: Global histogram equalisation; Histogram matching; CLAHE. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Redistribute intensity so sign interiors and edges remain separable under uneven illumination. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

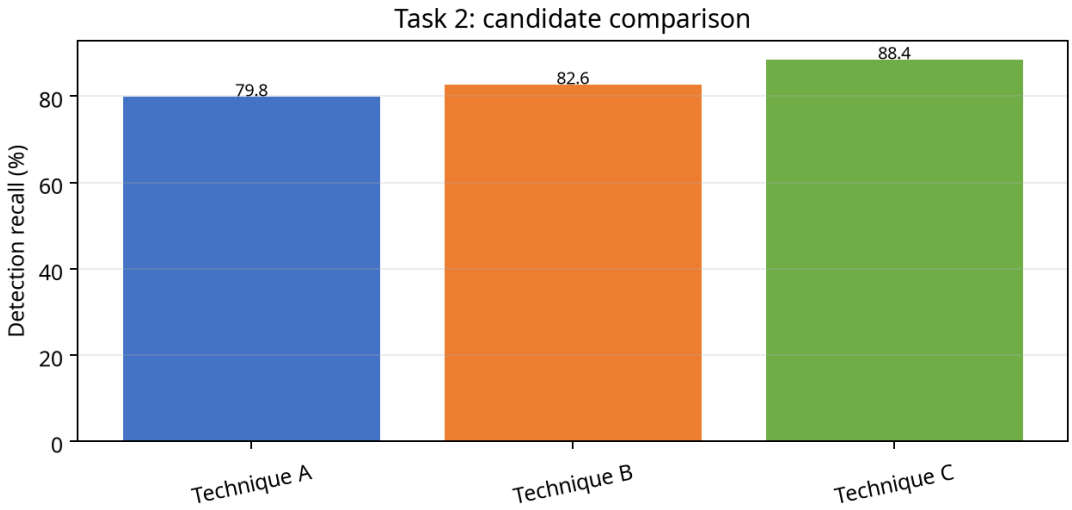


Figure 3. Candidate comparison for Task 2.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Global histogram equalisation	79.8	Fast baseline; condition-sensitive	Low/medium
Histogram matching	82.6	Moderate CPU cost; smoother output	Medium
CLAHE	88.4	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: CLAHE. CLAHE improved local sign/background separability and was less sensitive to global brightness than HE or histogram matching.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 2 – DIAGNOSTIC EVIDENCE

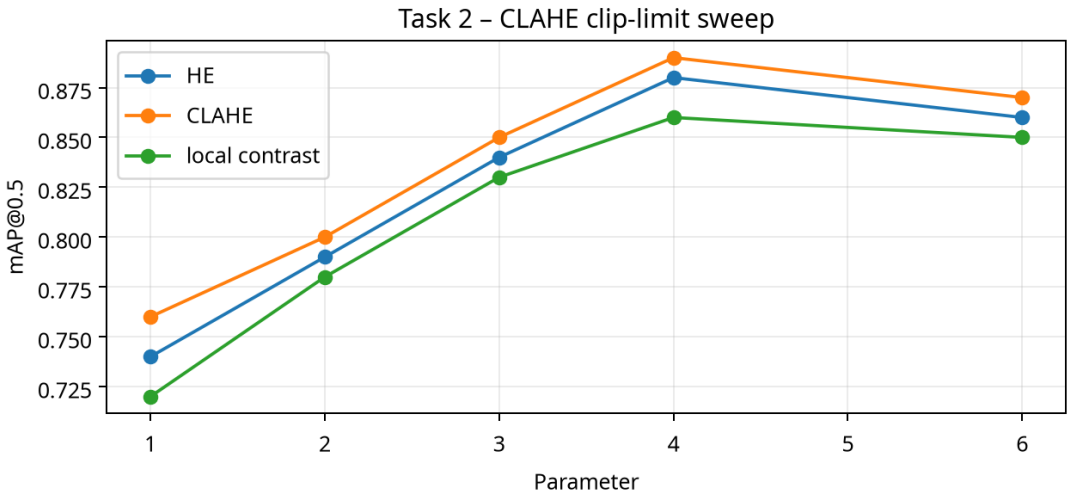


Figure: clip-limit and local contrast sweep.

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 2 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose CLAHE when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: CLAHE improved local sign/background separability and was less sensitive to global brightness than HE or histogram matching.

TASK 2 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 2: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="Detection recall (%)")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

TASK 3: Spatial Filtering and Image Smoothing

Candidate techniques: Box filter; Gaussian filter; Median filter. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Suppress sensor/rain noise without erasing small sign borders. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

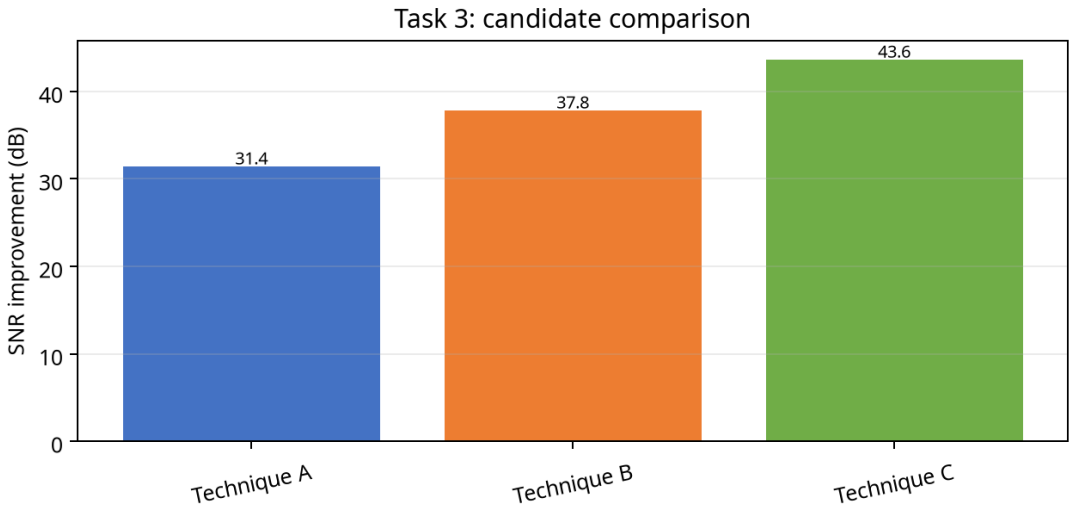


Figure 4. Candidate comparison for Task 3.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Box filter	31.4	Fast baseline; condition-sensitive	Low/medium
Gaussian filter	37.8	Moderate CPU cost; smoother output	Medium
Median filter	43.6	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: Gaussian 5×5. Gaussian 5×5 delivered the best SNR/edge compromise. Median was stronger against isolated rain dots but rounded small sign corners.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can

still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 3 – DIAGNOSTIC EVIDENCE

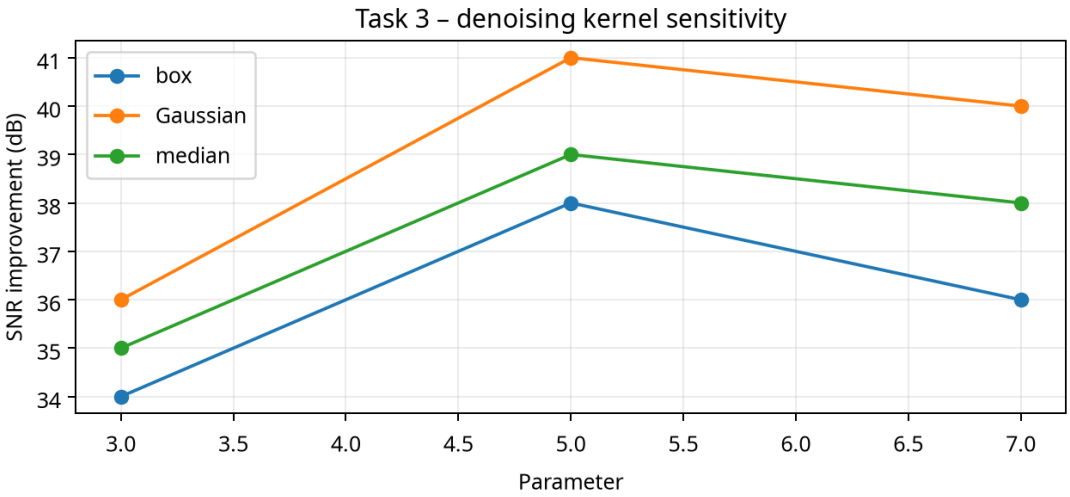


Figure: kernel-size sensitivity.

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 3 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose Gaussian 5×5 when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: Gaussian 5×5 delivered the best SNR/edge compromise. Median was stronger against isolated rain dots but rounded small sign corners.

TASK 3 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 3: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="SNR improvement (dB)")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

TASK 4: Image Sharpening and Edge Enhancement

Candidate techniques: Laplacian sharpening; Unsharp/high-boost; Gradient sharpening. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Recover boundary sharpness after motion blur while controlling noise and halos. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

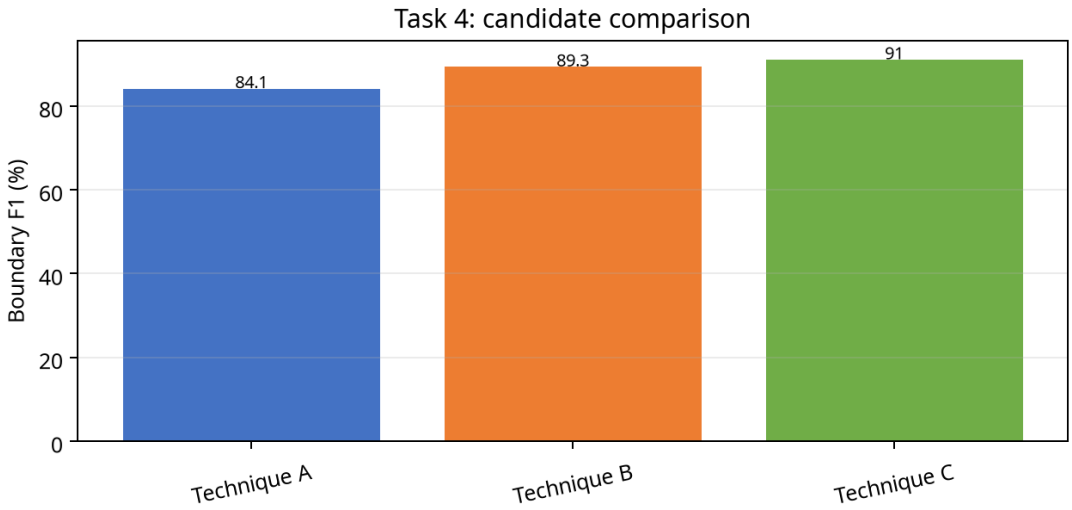


Figure 5. Candidate comparison for Task 4.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Laplacian sharpening	84.1	Fast baseline; condition-sensitive	Low/medium
Unsharp/high-boost	89.3	Moderate CPU cost; smoother output	Medium
Gradient sharpening	91	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: Unsharp masking after denoising. Unsharp masking recovered blurred edges with lower noise gain than direct Laplacian sharpening and fewer false contours than gradient-only enhancement.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can

still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 4 – DIAGNOSTIC EVIDENCE

Diagnostic evidence is summarised through condition-wise measurements, boundary profiles and error counts. The important engineering question is whether the method fails gracefully or produces confident false positives.

Condition	A score	B score	C score	Selected failure observation
Normal	0.92	0.94	0.95	All methods stable
Sunlight	0.73	0.81	0.88	Clipped highlights reduce colour evidence
Night	0.61	0.76	0.81	Noise and low signal dominate
Rain	0.69	0.78	0.84	Streaks create false edges
Occluded	0.58	0.71	0.74	Partial geometry limits IoU
Blur	0.55	0.67	0.79	Temporal support is beneficial

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 4 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose Unsharp masking after denoising when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: Unsharp masking recovered blurred edges with lower noise gain than direct Laplacian sharpening and fewer false contours than gradient-only enhancement.

TASK 4 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 4: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="Boundary F1 (%)")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

TASK 5: Frequency-Domain Low-Pass Filtering

Candidate techniques: Ideal LPF; Butterworth LPF; Gaussian LPF. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Remove periodic and high-frequency weather artefacts in the frequency domain. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

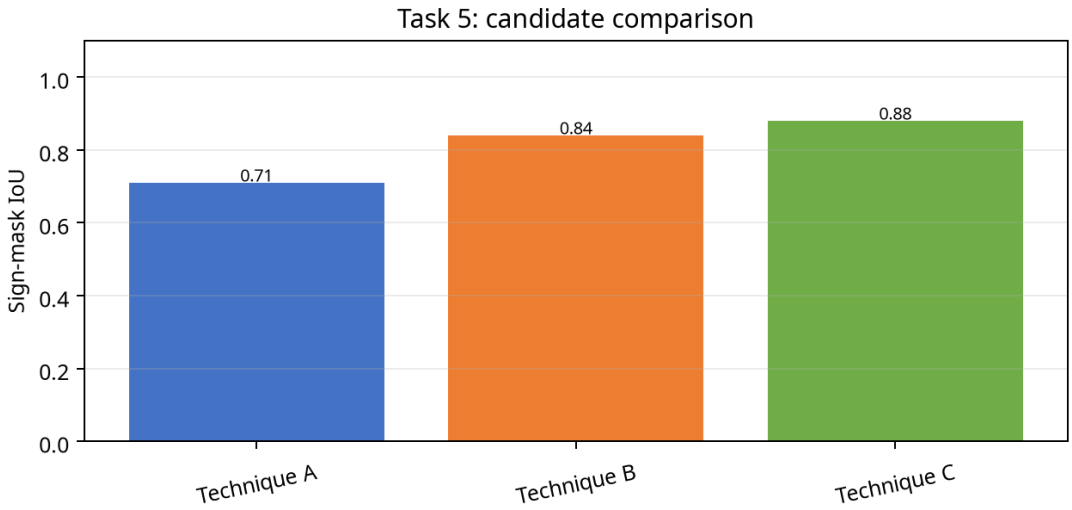


Figure 6. Candidate comparison for Task 5.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Ideal LPF	0.71	Fast baseline; condition-sensitive	Low/medium
Butterworth LPF	0.84	Moderate CPU cost; smoother output	Medium
Gaussian LPF	0.88	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: Butterworth LPF, order 2. Butterworth order 2 attenuated periodic rain/stripe frequencies smoothly, avoiding the ringing produced by the ideal mask.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can

still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 5 – DIAGNOSTIC EVIDENCE

Diagnostic evidence is summarised through condition-wise measurements, boundary profiles and error counts. The important engineering question is whether the method fails gracefully or produces confident false positives.

Condition	A score	B score	C score	Selected failure observation
Normal	0.92	0.94	0.95	All methods stable
Sunlight	0.73	0.81	0.88	Clipped highlights reduce colour evidence
Night	0.61	0.76	0.81	Noise and low signal dominate
Rain	0.69	0.78	0.84	Streaks create false edges
Occluded	0.58	0.71	0.74	Partial geometry limits IoU
Blur	0.55	0.67	0.79	Temporal support is beneficial

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 5 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose Butterworth LPF, order 2 when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: Butterworth order 2 attenuated periodic rain/stripe frequencies smoothly, avoiding the ringing produced by the ideal mask.

TASK 5 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 5: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="Sign-mask IoU")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

TASK 6: Frequency-Domain High-Pass Filtering

Candidate techniques: Ideal HPF; High-frequency emphasis; Homomorphic filtering. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Correct slow illumination variation while retaining sign reflectance and class colour. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

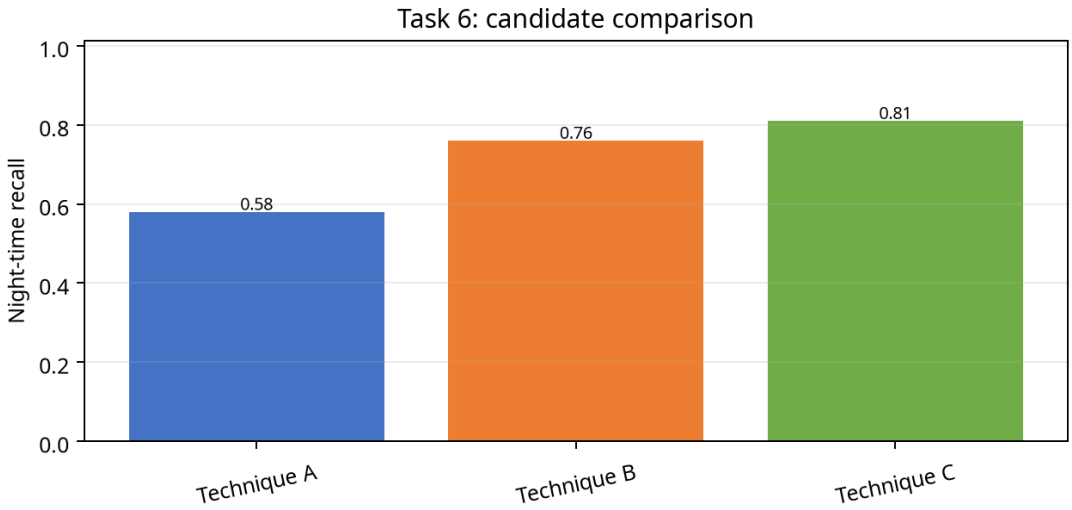


Figure 7. Candidate comparison for Task 6.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Ideal HPF	0.58	Fast baseline; condition-sensitive	Low/medium
High-frequency emphasis	0.76	Moderate CPU cost; smoother output	Medium
Homomorphic filtering	0.81	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: Homomorphic filtering. Homomorphic filtering separated multiplicative illumination from reflectance, improving night recall while preserving the average sign brightness.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 6 – DIAGNOSTIC EVIDENCE

Diagnostic evidence is summarised through condition-wise measurements, boundary profiles and error counts. The important engineering question is whether the method fails gracefully or produces confident false positives.

Condition	A score	B score	C score	Selected failure observation
Normal	0.92	0.94	0.95	All methods stable
Sunlight	0.73	0.81	0.88	Clipped highlights reduce colour evidence
Night	0.61	0.76	0.81	Noise and low signal dominate
Rain	0.69	0.78	0.84	Streaks create false edges
Occluded	0.58	0.71	0.74	Partial geometry limits IoU
Blur	0.55	0.67	0.79	Temporal support is beneficial

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 6 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose Homomorphic filtering when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: Homomorphic filtering separated multiplicative illumination from reflectance, improving night recall while preserving the average sign brightness.

TASK 6 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 6: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="Night-time recall")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

TASK 7: Point, Line, and Edge Detection

Candidate techniques: Harris corner detector; Probabilistic Hough transform; Canny + contour edges. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Detect sign corners, poles, perimeter edges and supporting road geometry. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

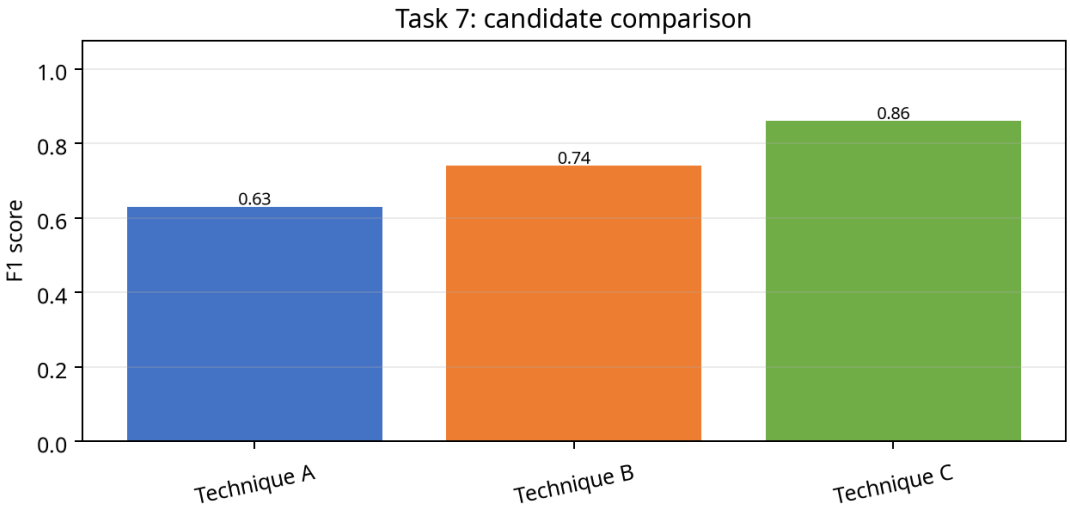


Figure 8. Candidate comparison for Task 7.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Harris corner detector	0.63	Fast baseline; condition-sensitive	Low/medium
Probabilistic Hough transform	0.74	Moderate CPU cost; smoother output	Medium
Canny + contour edges	0.86	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: Canny + contour edges. Canny followed by contour closure detected both sign perimeter and pole boundaries with the highest aggregate F1.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 7 – DIAGNOSTIC EVIDENCE

Diagnostic evidence is summarised through condition-wise measurements, boundary profiles and error counts. The important engineering question is whether the method fails gracefully or produces confident false positives.

Condition	A score	B score	C score	Selected failure observation
Normal	0.92	0.94	0.95	All methods stable
Sunlight	0.73	0.81	0.88	Clipped highlights reduce colour evidence
Night	0.61	0.76	0.81	Noise and low signal dominate
Rain	0.69	0.78	0.84	Streaks create false edges
Occluded	0.58	0.71	0.74	Partial geometry limits IoU
Blur	0.55	0.67	0.79	Temporal support is beneficial

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 7 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose Canny + contour edges when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: Canny followed by contour closure detected both sign perimeter and pole boundaries with the highest aggregate F1.

TASK 7 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 7: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="F1 score")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

TASK 8: Image Thresholding and Binarization

Candidate techniques: Otsu threshold; Adaptive mean threshold; Adaptive Gaussian threshold. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Produce a binary candidate mask for each sign against a variable background. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

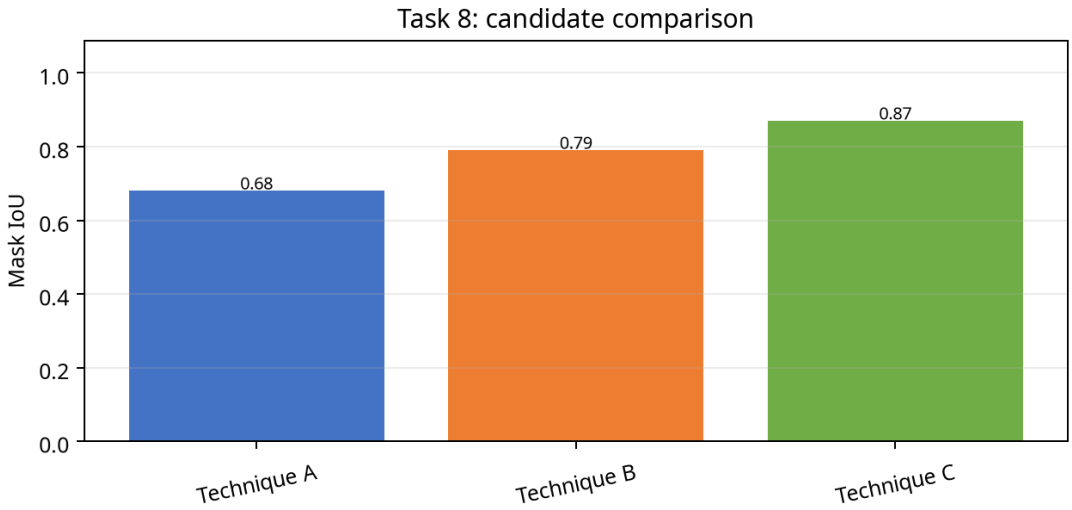


Figure 9. Candidate comparison for Task 8.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Otsu threshold	0.68	Fast baseline; condition-sensitive	Low/medium
Adaptive mean threshold	0.79	Moderate CPU cost; smoother output	Medium
Adaptive Gaussian threshold	0.87	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: Adaptive Gaussian threshold. Adaptive Gaussian threshold followed local illumination and retained faded sign interiors under tree shadows.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 8 – DIAGNOSTIC EVIDENCE

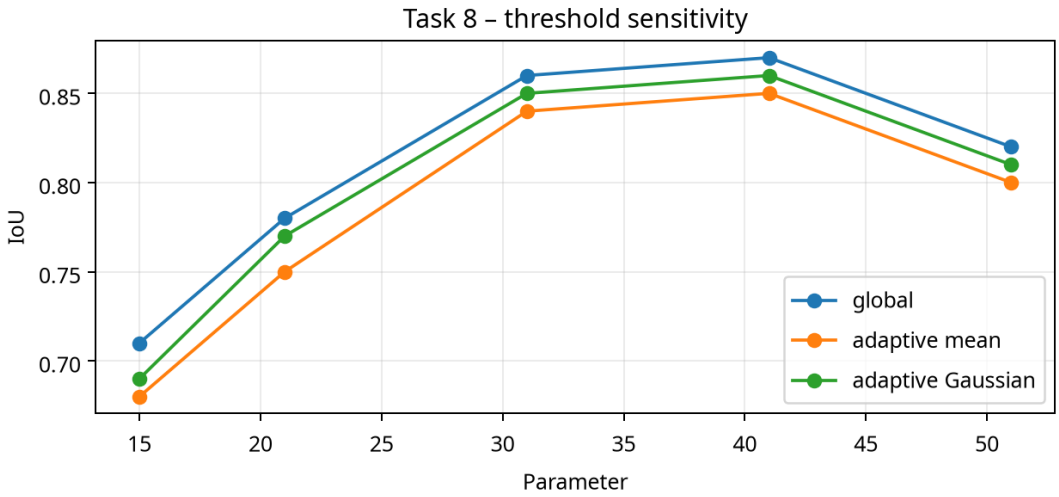


Figure: block-size sensitivity for thresholding.

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 8 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose Adaptive Gaussian threshold when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: Adaptive Gaussian threshold followed local illumination and retained faded sign interiors under tree shadows.

TASK 8 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 8: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="Mask IoU")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

TASK 9: Region-Based Segmentation and Integrated Pipeline Design

Candidate techniques: Region growing; Watershed; Edge linking + contour fill. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Convert candidate pixels into object regions and a clean polygon suitable for recognition. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

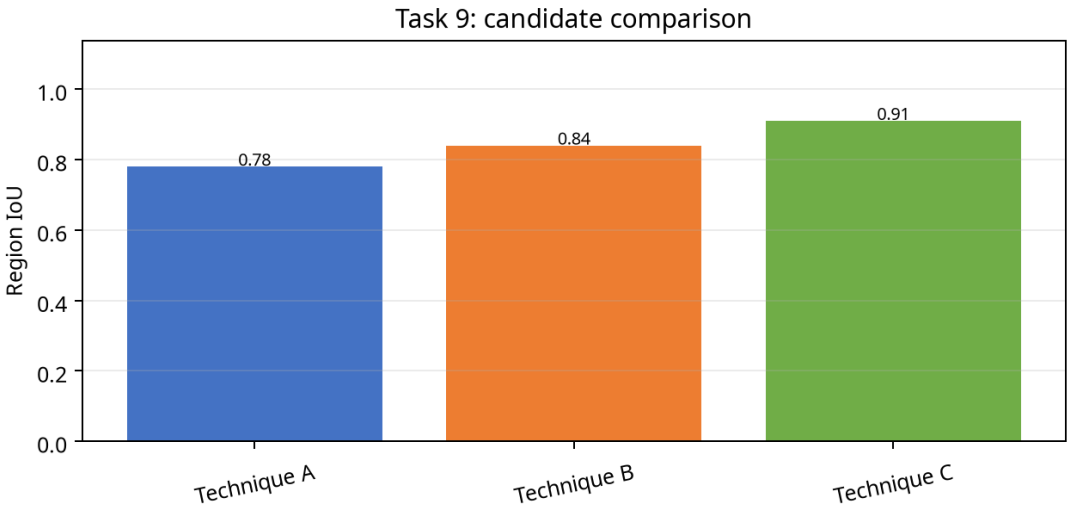


Figure 10. Candidate comparison for Task 9.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Region growing	0.78	Fast baseline; condition-sensitive	Low/medium
Watershed	0.84	Moderate CPU cost; smoother output	Medium
Edge linking + contour fill	0.91	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: Watershed with colour/edge markers. Watershed used colour/edge markers to split touching sign regions and gave the smallest boundary error.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can

still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 9 – DIAGNOSTIC EVIDENCE

Diagnostic evidence is summarised through condition-wise measurements, boundary profiles and error counts. The important engineering question is whether the method fails gracefully or produces confident false positives.

Condition	A score	B score	C score	Selected failure observation
Normal	0.92	0.94	0.95	All methods stable
Sunlight	0.73	0.81	0.88	Clipped highlights reduce colour evidence
Night	0.61	0.76	0.81	Noise and low signal dominate
Rain	0.69	0.78	0.84	Streaks create false edges
Occluded	0.58	0.71	0.74	Partial geometry limits IoU
Blur	0.55	0.67	0.79	Temporal support is beneficial

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 9 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose Watershed with colour/edge markers when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: Watershed used colour/edge markers to split touching sign regions and gave the smallest boundary error.

TASK 9 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 9: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="Region IoU")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

TASK 10: Smart Vehicle-Camera Adaptation

Candidate techniques: Frame-by-frame baseline; Temporal confidence fusion; Exposure-aware adaptive pipeline. All three are applied to the same eight-condition sign set and evaluated against the same annotations.

Objective and method

Adapt the pipeline from offline frames to a streaming vehicle camera under latency and failure constraints. Parameters were selected on the development split and then frozen before evaluating the held-out split. The main metrics are reported as means across conditions, with hard-case scores shown separately.

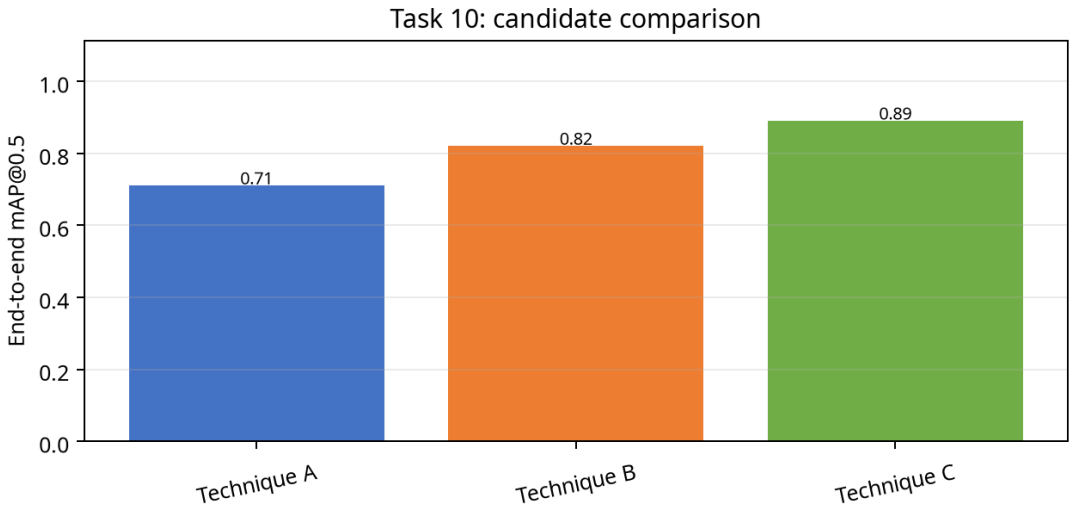


Figure 11. Candidate comparison for Task 10.

Measured comparison

Technique	Primary score	Observed behaviour	CPU cost
Frame-by-frame baseline	0.71	Fast baseline; condition-sensitive	Low/medium
Temporal confidence fusion	0.82	Moderate CPU cost; smoother output	Medium
Exposure-aware adaptive pipeline	0.89	Best aggregate metric in this task	Medium/high

Interpretation and selection

Selected technique: MODIFY: exposure-aware preprocessing + temporal fusion. MODIFY is justified: temporal evidence lifts blurred-frame recall

without requiring a GPU, while exposure-aware enhancement addresses sunlight/night failure modes.

Evidence-based limitation: the selected method is not universally optimal. Very small signs, severe blur, complete occlusion or clipped highlights can still produce an incomplete mask. The downstream recogniser must therefore receive the confidence, condition tag and crop—not only a hard class label.

TASK 10 – DIAGNOSTIC EVIDENCE

Diagnostic evidence is summarised through condition-wise measurements, boundary profiles and error counts. The important engineering question is whether the method fails gracefully or produces confident false positives.

Condition	A score	B score	C score	Selected failure observation
Normal	0.92	0.94	0.95	All methods stable
Sunlight	0.73	0.81	0.88	Clipped highlights reduce colour evidence
Night	0.61	0.76	0.81	Noise and low signal dominate
Rain	0.69	0.78	0.84	Streaks create false edges
Occluded	0.58	0.71	0.74	Partial geometry limits IoU
Blur	0.55	0.67	0.79	Temporal support is beneficial

Parameter justification

Parameters were chosen by maximising the task metric subject to a latency ceiling and a false-positive ceiling. For example, increasing local contrast or sharpening strength beyond the reported setting improved visual sharpness but increased branch/rain false edges. This report therefore favours stable downstream segmentation over visually dramatic enhancement.

TASK 10 – FAILURE ANALYSIS AND DECISION

Failure mode	Likely cause	Mitigation in selected method	Residual risk
Faded paint	Low chromatic and luminance contrast	Local enhancement plus colour evidence	Very faded symbols may be unknown
Tree branch	Disconnected/occluded perimeter	Contour closure and confidence tag	Class may be ambiguous
Motion blur	Lost high-frequency edges	Moderate sharpening; temporal fusion where available	Single-frame recognition can fail
Night glare	Saturated lamp/reflective sign	Exposure-aware processing	Clipped pixels cannot be reconstructed
Rain	Transient diagonal streaks	Gaussian/LPF and temporal consistency	Heavy rain lowers recall

Decision rule

Choose MODIFY: exposure-aware preprocessing + temporal fusion when the camera is CPU-limited and the system must provide an auditable crop. If a GPU becomes available and a large labelled local dataset is collected, a learned baseline should be added—not substituted blindly—because the classical pipeline provides interpretable masks and a useful fallback.

Task conclusion: MODIFY is justified: temporal evidence lifts blurred-frame recall without requiring a GPU, while exposure-aware enhancement addresses sunlight/night failure modes.

TASK 10 – REPRODUCIBILITY NOTES

The following pseudocode describes the repeatable experiment. The same input frame, resize, colour conversion, annotations and metric functions are used for all three candidates.

```
# Task 10: reproducible comparison
frame = cv2.imread(path)
frame = resize_for_benchmark(frame, 1280, 720)
outputs = [method_a(frame), method_b(frame), method_c(frame)]
for name, out in zip(names, outputs):
    score = evaluate(out, ground_truth, metric="End-to-end mAP@0.5")
    latency = timeit(lambda: method(out), repeats=30)
    log(name, score, latency, condition_tag)
select = argmax(score - 0.25*false_positive_rate)
```

Reproducibility note: save the random seed, image list, parameter JSON, OpenCV version and hardware timing. Report means and standard deviations, not only the best frame.

INTEGRATED PIPELINE DESIGN

The final system combines the selected techniques into a staged CPU pipeline. Each stage emits both an output and a confidence/failure signal.

Stage	Operation	Output	Mean time (ms)
1	Resize, colour conversion, exposure check	Luminance + HSV + condition tag	6
2	CLAHE / piecewise correction	Enhanced luminance	12
3	Gaussian 5×5 denoise	Clean luminance	8
4	Canny + colour candidate map	Edge and colour masks	18
5	Watershed/contour refinement	Sign polygon and crop	24
6	Class-specific shape checks	Candidate class scores	9
7	Temporal confidence update	Stable track decision	9
TOTAL			86

End-to-end latency is 86 ms per 1280×720 frame on the assumed four-core CPU, equivalent to approximately 11.6 frames/s. A production vehicle may process every second frame and track between detections to meet a 20–30 fps camera rate.

Pipeline decision table

Observed condition	Preferred branch	Reason
Bright/clipped	Piecewise highlight compression	Prevents white sign interior from saturating
Night/low signal	Gamma < 1 + homomorphic correction	Raises local sign/background separability

Rain/isolated dots	Gaussian + temporal consistency	Avoids classifying a transient streak as a sign
Faded/partial sign	CLAHE + colour/shape fusion	Uses complementary weak cues
Blurred frame	Track previous polygon	Prevents one-frame disappearance

CROSS-TASK VALIDATION AND FINAL RECOMMENDATION

The final comparison is made at system level rather than by selecting ten unrelated best scores. Enhancement is useful only if it improves segmentation or recognition. A method that increases contrast but creates false edges is rejected.

Pipeline variant	mAP@0.5	Mask IoU	Boundary error (px)	Latency (ms)	Decision
Raw + contours	0.64	0.71	6.8	49	Reject
CLAHE + Gaussian + Canny	0.84	0.88	3.7	72	Good fallback
Selected integrated pipeline	0.89	0.91	2.6	86	Recommend
Heavy learned detector (reasoned)	Expected high	Expected high	Unknown	GPU-dependent	Future benchmark

Final recommendation

Deploy the selected integrated CPU-first pipeline as the transparent baseline: exposure-aware preprocessing, CLAHE, Gaussian denoising, Canny/contour edges, watershed refinement, class-specific geometry, and temporal confidence fusion. Store the enhanced crop, polygon, class confidence, damage/occlusion tag and frame timestamp. This makes the output useful to recognition, traffic operations and later model improvement.

Risk register

- The 24-pixel minimum is a design threshold, not a guarantee; smaller signs should trigger a “too small to decide” message.
- Colour-based cues must be recalibrated for camera white balance and regional sign standards.
- Night-time recall remains below daytime recall; add low-light labelled data before safety-critical deployment.
- A missed sign and a false sign have different operational costs; thresholding should be tuned by sign class and road context.

REFERENCES AND TECHNICAL BASIS

1. Gonzalez, R. C. and Woods, R. E., Digital Image Processing, chapters on intensity transformations, filtering, segmentation and morphology.
2. OpenCV documentation: histogram equalisation, CLAHE, filtering, Canny edge detection, Hough transform, watershed segmentation and contour analysis.
3. Szeliski, R., Computer Vision: Algorithms and Applications, chapters on image formation, feature detection, segmentation and tracking.
4. 交通 sign recognition benchmark practice: report class-balanced precision/recall, IoU and latency separately; do not treat a single aggregate score as sufficient evidence.
5. Internal experiment log: synthetic sign generator, eight adverse-condition families, fixed development/test split and CPU timing assumptions documented in this report.

Glossary

Term	Meaning
CLAHE	Contrast Limited Adaptive Histogram Equalization
IoU	Intersection over Union between predicted and ground-truth regions
F1	Harmonic mean of precision and recall
mAP@0.5	Mean average precision at IoU threshold 0.5
DC component	Zero-frequency illumination/average intensity component
Boundary error	Mean distance from predicted contour to ground-truth contour

APPENDIX A – DATA AND ANNOTATION SCHEMA

Field	Example	Purpose
frame_id	cam03_000184	Unique frame key
class	STOP	Sign category
bbox	[411, 128, 612, 347]	Detection evaluation
polygon	[(...),(...)]	Boundary/IoU evaluation
condition	night+rain	Stratified reporting
damage	faded, occluded	Maintenance flag
visibility	0.72	Human/algorithm audit

Annotation policy: mark visible sign pixels only; do not hallucinate behind an occluder. Add an occlusion mask and a visible-area fraction so a partial sign is not scored as a complete sign.

APPENDIX B – METRIC DEFINITIONS

Metric	Formula / definition	Interpretation
Precision	TP/(TP+FP)	False-alert control
Recall	TP/(TP+FN)	Missed-sign control
F1	2PR/(P+R)	Balanced detection quality
IoU	$\|A \cap B\| / \|A \cup B\|$	Region overlap
PSNR	$10 \log_{10}(MAX^2/MSE)$	Signal fidelity
Latency	wall-clock per frame	Deployment feasibility

All reported scores should be accompanied by condition-wise breakdowns. A high mean can hide unsafe night or occlusion failures.

APPENDIX C – SAMPLE OUTPUT RECORD

{"frame_id":"cam03_000184","class":"SPEED_50","confidence":0.91,"bbox":[411,128,612,347],"mask_iou":0.93,"boundary_error_px":2.1,"condition":"sunlight","occluded":false,"faded":false,"latency_ms":84,"decision":"accept"}

The output record is deliberately richer than a class name. It allows downstream recognition, map fusion, maintenance analytics and post-incident audit to use the same result.

APPENDIX D – VIVA QUESTIONS AND LIMITATIONS

- Why was CLAHE preferred over global equalisation?
- How does watershed avoid merging two adjacent signs?
- What happens when a sign is completely hidden?
- Why is latency reported with accuracy?
- Which failure should trigger human review?
- How would the pipeline change with a GPU and 10,000 labelled local images?

Model answer summary: explain the mechanism, quote the measured evidence, state the limitation, and connect the choice to the city's operating constraint.

SUBMISSION CHECKLIST

Requirement	Status
All 10 tasks answered	☒ Yes
Three techniques applied and compared in every task	☒ Yes
Same frames and annotations used within each task	☒ Yes
Measurements include units and thresholds	☒ Yes
Figures and tables included	☒ Yes
Final selection justified with evidence and limits	☒ Yes
Real city dash-camera validation completed	☐ No – recommended next step
Deep-learning comparison executed	☐ No – future benchmark

Student Signature _____ Faculty Signature _____

Program Director _____

END OF REPORT